

MÁRIO AFONSO

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SUMMARY

Systems Engineering student with a strong foundation in systems programming, Linux and software development. Previous professional experience includes critical datacenter operations and infrastructure monitoring in high-availability environments, as well as freelance development of web applications and operational dashboards using TypeScript, React, Next.js and Node.js. Academic projects include IoT and edge-computing systems built with MQTT, EdgeX Foundry and machine learning techniques for predictive maintenance. Currently completing the 42 Porto Software Engineering Program (Milestone 3) while pursuing a Master's degree in Computer Engineering at Universidade Portucalense.

TECHNICAL SKILLS

Languages: C, Python, TypeScript, Bash

Software Engineering: Algorithms, Data Structures, Object-Oriented Programming, Systems Programming, Memory Management

Frontend: React, Next.js, TailwindCSS, Framer Motion

Backend: Node.js, Express

Databases: MongoDB, MySQL, InfluxDB

Infrastructure: Linux, Docker, Docker Compose, Git, Zabbix

IoT & Edge: EdgeX Foundry, MQTT, Grafana, Sensor Networks

PROFESSIONAL EXPERIENCE

ArduMaker, Remote: Freelance Full-Stack Developer

2025 — Present

- Designed and developed operational dashboards for chatbot analytics and vendor management used across multiple locations.
- Built internal agricultural management platform for monitoring field operations and workflow coordination.
- Delivered production systems used by internal teams for operational monitoring and decision-making.

CISP (Integrated Public Security Center), Benguela, Angola: Data Center Technician

Apr 2020 – Aug 2023

- Managed and monitored critical IT infrastructure using Linux, Zabbix, and proprietary datacenter platforms.
- Performed system maintenance, troubleshooting and uptime optimisation in high-availability environments.

Angola Telecom, Benguela, Angola: Technical Assistant (Internship)

May 2019 – Aug 2019

- Assisted in network operations, equipment configuration and incident resolution for telecommunications infrastructure.

PROJECTS

42 Porto — Systems Programming Curriculum

2025 — Present

Rigorous peer-reviewed software engineering programme focused on C, algorithms and systems programming. Currently at Milestone 3, progressing to Milestone 4.

- Milestone 1:** *Libft* (C standard library, 40+ functions), *ft_printf* (variadic argument parsing), *get_next_line* (buffered line reading via file descriptors), *Born2beRoot* (Linux server hardening),
- Milestone 2:** *push_swap* (dual-stack sorting optimisation), *Python Modules* (OOP, iterators, decorators, generators), *a-mazing* (maze generator and solver using graph traversal).
- Milestone 3:** *Fly-in (Drone Traffic Simulation Platform)*, Event-driven drone traffic simulator developed in Python with graphical visualisation and configurable environments, developed a configurable drone traffic simulation engine supporting path planning, congestion handling and priority rules, built graphical visualisation tools for real-time monitoring and debugging.
- All projects peer-reviewed under zero-tolerance policy for memory leaks and undefined behaviour.

Edge-Based Failure Prediction System

2024

IoT and machine learning platform for predictive maintenance using edge computing (Academic — UPT).

- Built edge IoT network using EdgeX Foundry and MQTT;
- Applied ML to predict system failures from sensor data.
- Implemented master-slave architecture in Python for coordinated sensor data acquisition across multiple edge devices.

- Developed real-time monitoring dashboard with Next.js for live sensor metrics and system health visualisation.

DigitalMTX – Chatbot Management Dashboard

2025

Admin dashboard for managing chatbot interactions, vendors and operational metrics across multiple locations. Stack: Next.js, TailwindCSS, TypeScript.

- Developed modules for vendor management, chat history and real-time interaction analytics.
- Integrated frontend with backend APIs for live chatbot data visualisation.

EDUCATION

MSc Computer Engineering and Computing

2024 — Present

Universidade Portucalense

Coursework: Cloud Infrastructure, Computer Networks, Databases, IoT Integration, Data Science, and AI-Driven Collaborative Projects

Software Engineering Program — Algorithms, Data Structures, Systems Programming, C, Python

2025 — Present

42 Porto

Coursework: Algorithms, Data Structures, Computer Networks, Databases & Systems Programming, Collaborative Projects

Bachelor’s Degree in Computer Engineering

2016 — 2021

Instituto Superior Politécnico Kalandula de Angola

Coursework: Software Engineering, Distributed Systems, Databases, Computer Networks, Artificial Intelligence, Cloud & IoT, Information Security, Project Management, Business Management

AVAILABILITY & ELIGIBILITY

- EU work permit
- Already living in Porto and open to Lisbon or Coimbra (relocation ready)